Invited Paper

Field, temperature, and anchoring induced inhomogeneous index distribution of a nano-dispersed nematic liquid crystal near the Freedericksz transition: Monte Carlo studies

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ABSTRACT

Monte Carlo studies of the field induced complex refractive index changes in nano-dispersed nematic liquid crystals exhibiting negative - positive refractive indices[1,2] have been performed for various cases of applied field strengths,[3] anchoring profiles and temperatures over a broad spectral regime. The resultant induced spatially inhomogeneous molecular order and the corresponding metamaterial properties are obtained for various wavelengths, applied field strengths, anchoring forces and temperatures below and above the Freedericksz transition. In general, the director axis reorientation and the resultant refractive index distribution are spatially inhomogeneous, even under an uniform applied field. The detailed computation have identified parameter sets for obtaining negative index of refraction and maximal index modulations that can be more than an order of magnitude larger than those obtainable from pure NLC systems.

Keywords: negative index of refraction, metamaterials, nematic liquid crystals, Monte Carlo

1. INTRODUCTION

Since the pioneering work of Veselago,[4] metamaterials having negative refractive indices have been proposed and demonstrated in microwave – optical region. Tunability of the refractive index of the metamaterials is an important requirement for optical applications. In the case of materials based on liquid crystals host the tunability is enabled by electrical and/or optical field controlled birefringence of the nematic liquid crystals constituents.[1,2,5–10] Nonlocal inhomogeneous nematic responses are particularly important to obtain accurate quantitative description of the index modulation. Furthermore, in cases where non-uniform fields are applied, for example, in photorefractivity-mediated optical wave mixing processes,[10] these approaches often produce new insights, and unexpected and potentially useful effects.

The aim of this paper is to review briefly our recent studies related to tuning of negative-zero-positive index of refraction using inhomogeneous electric field and anchoring forces, as well as to discuss the role of temperature. For this purpose, we apply Monte Carlo simulation techniques, which have been demonstrated to be well–suited for modelling such NLC responses.[11,12]

2. NANO-SPHERE DISPERSED LIQUID CRYSTAL

The presentation in this Section is based on papers.[3,13] In the formalism introduced by Ref.,[1,7] electromagnetic wave scattering from a single coated sphere is calculated using Mie theory[14] while the effective scattering from an ensemble of coated spheres dispersed in NLC matrix is analyzed in effective medium theory using Maxwell Garnet mixing rule.[15,16] The effective refractive index of NDLC depends on the parameters of its components: coated nonmagnetic spheres, and host NLC system (Fig. 1). Fig. 2 shows the schematic picture of a coated nano–sphere. The core with radius r_1 is made from polaritonic material (such as $LiTaO_3$) with relative permittivity

$$\epsilon_1(\omega) = \epsilon_1(\infty)\left(1 + \frac{\omega_L^2 - \omega_T^2}{\omega_T^2 - \omega^2 - i\omega\gamma_1}\right). \tag{1}$$

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Liquid Crystals XIV, edited by Iam Choon Khoo, Proc. of SPIE Vol. 7775
77750K · © 2010 SPIE · CCC code: 0277-786X/10/$18 · doi: 10.1117/12.860126

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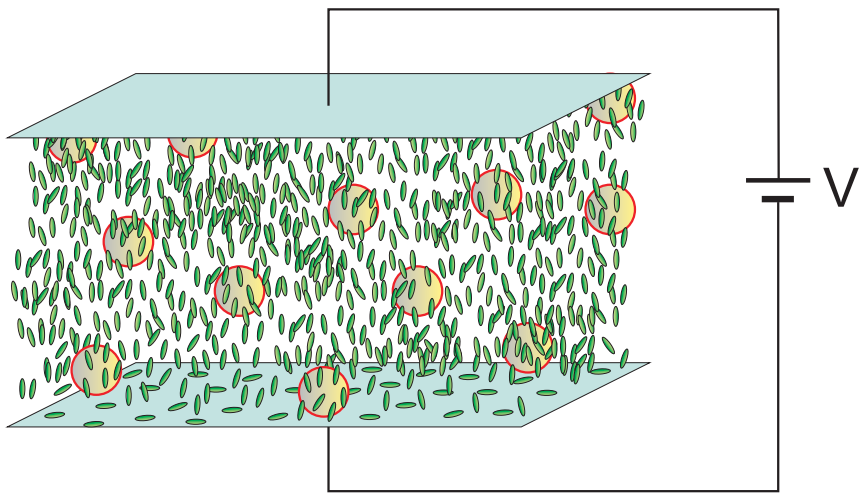

Figure 1. Nanosphere-doped liquid crystal cell.

$\epsilon_1(\infty)$ denotes the high-frequency dielectric constant, γ_1 – damping coefficient, ω_T and ω_L – the transverse and longitudinal phonon frequencies, respectively. Bulk cavity resonance in the core is responsible for negative effective permeability of the assembly of spheres at particular electromagnetic wavelengths. The relative permittivity of the semiconductor shell with outer radius r_2 is described by a Drude model:

$$\epsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma_2}, \tag{2}$$

where ω_p is plasma frequency and γ_2 – the damping coefficient. The shell is responsible for negative effective permittivity, with ω_p close to frequency for which negative index will occur. Its value is chosen to provide both negative effective permittivity and permeability.

Permittivity of the liquid crystal host for extraordinary polarized wave depends on angle between directions of incident light wave vector and the director axis of NLC:[1,2,10]

$$\epsilon_3(\theta) = \frac{\epsilon_\parallel \epsilon_\perp}{\epsilon_\parallel \cos^2(\theta) + \epsilon_\perp \sin^2(\theta)}, \tag{3}$$

where $\epsilon_\parallel$ and $\epsilon_\perp$ are the permittivities for light polarized respectively parallel and perpendicular to the director axis. In pure NLC host $k_3 = \sqrt{\epsilon_3}\, k_0$.

The effective permittivity and permeability of NDLC, calculated in effective medium theory in dipolar (electric and magnetic) approximations are given by formulas:[1]

$$\epsilon_{eff} = \epsilon_3 \left(\frac{k_3^3 + 4\pi i N a_1}{k_3^3 - 2\pi i N a_1} \right), \tag{4}$$

$$\mu_{eff} = \frac{k_3^3 + 4\pi i N b_1}{k_3^3 - 2\pi i N b_1}. \tag{5}$$

More information, in particular on coefficients a_1 and b_1, can be found in.[14] These expressions are valid only for small number density $N = 3f/4\pi r_2^3$ of the nano-spheres (f is the filling factor) and in long-wavelength limit $2\pi/k_3 \gg r_2$. The effective index of refraction is

$$n_{eff} = \pm\sqrt{\epsilon_{eff}\, \mu_{eff}}, \tag{6}$$

where we have used the root corresponding to a positive imaginary part. Based on homogeneous–response formalism, Khoo et al. have shown[1] that for some values of the parameters of the model there is a range of frequencies ω and angles θ for which the real part of effective refractive index is negative.

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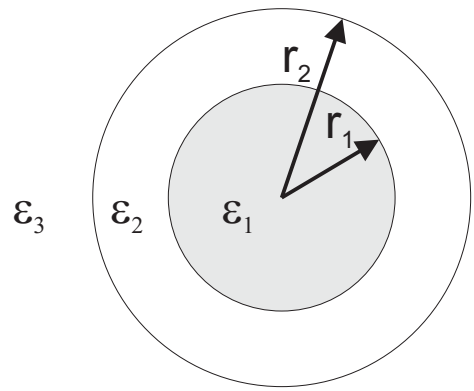

Figure 2. Coated nano-sphere.

3. MONTE CARLO SIMULATIONS AND NDLC MODEL

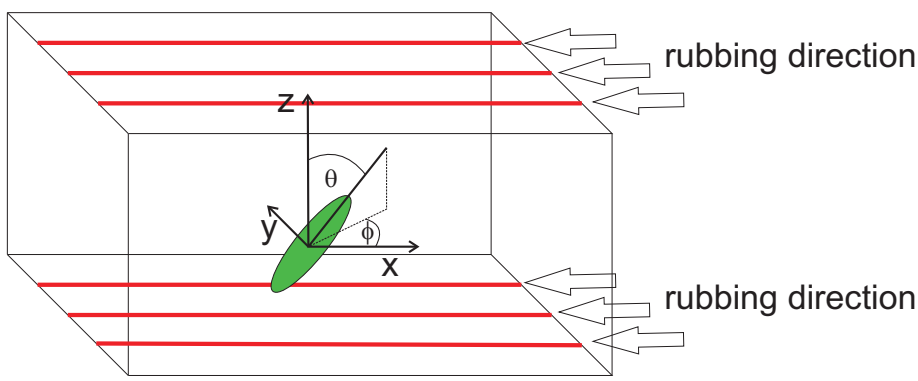

Figure 3. Geometry of the system and orientation of LC molecule.

Monte CArlo simulation of the system was decribed in details in paper.[3] The NLC molecules, represented by 1D rods, occupy sites of a $n_x \times n_y \times n_z$ rectangular lattice. Translational motions of the center of mass of the molecules are prohibited. An orientation of a long axis of a NLC molecule in an external coordinate system is described by a pair of angles ϕ, θ (Fig. 3). The angle ϕ (twist) describes an azimuth and angle θ (tilt) is a polar angle calculated from z-axis, along which the electromagnetic wave propagates with polarization along x-axis. We apply rubbing along x-axis. The molecules lie (and fluctuate) in the $x-z$ plane. Along the x and y-direction the periodic boundary conditions are applied, while the molecules in the lowest (1) and the uppermost (n_z) planes are subjected to properly chosen anchoring forces due to their interaction with a substrate. Interactions are described by the Lebwohl-Lasher effective Hamiltonian[17] with Rapini-Papoular term.[18] Potential energy U (divided by k_BT, where k_B – Boltzmann constant, T – absolute temperature):

$$\frac{U}{k_BT} = -\frac{\xi}{k_BT} \sum_{<\vec{r},\vec{r}'>} P_2(\cos\beta(\vec{r},\,\vec{r}')) - \sum_{\vec{r}} \frac{E^2(\vec{r})}{k_BT} P_2(\cos\beta(\vec{r})) + \sum_{\vec{r}_W} \frac{\alpha(\vec{r}_W)}{k_BT} \sin^2\gamma(\vec{r}_W), \tag{7}$$

Here P_2 is the second-order Legendre polynomial. Parameter ξ is the coupling constant for NLC-NLC orientational interaction (in simulations we put $\xi = 25$), $E(\vec{r})$ is proportional to external electric field, $\alpha(\vec{r}_W)$ denotes the amplitude of anchoring forces acting on molecules located at the walls. $\beta(\vec{r},\,\vec{r}')$ denotes the relative angle between two molecules located at points $\vec{r}$, $\vec{r}'$. $\beta(\vec{r})$ is the angle a molecule located at point $\vec{r}$ makes with the electric field direction. $\gamma(\vec{r}_W)$ denotes the angle between long axis of molecule at the wall and a fixed rubbing

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direction (Fig. 3). Each molecule in the bulk is surrounded by the six nearest neighbors and interactions only with those neighbors are taken into account.

The equilibrium configurations of NLC molecules on a lattice were sampled using Metropolis Monte Carlo simulations.[19,20] The system was equilibrated during 3×10^4 MC steps (MCS) and then simulated during 2×10^5 MCS. From 2×10^5 equilibrium configurations generated during simulation 2×10^2 configurations (separated by 10^3 MCS) were used to calculate the averaged values of chosen physical properties. In the simulations we have used the parameters from Ref. 8: $r_1 = 0.13\ \mu\mathrm{m}, \mathrm{r}_2 = 0.143\ \mu\mathrm{m}, \mathrm{f} = 0.12, \epsilon(\infty) = 17, \omega_\mathrm{T}/2\pi = 240$ THz, $\omega_L/2\pi = 570$ THz, $\gamma_1/2\pi = 2.5$ THz, $\omega_\mathrm{p}/2\pi = 134$ THz and $\gamma_2 = \omega_p/60$.

4. HOMOGENEOUS ANCHORING STRENGTH AND SPATIALLY MODULATED ELECTRIC FIELD

Instead of global order[1] (Fig. 4 (a)) we have introduced model of the cell with anchoring forces and electric field (Fig. 4 (b)) and with additional spatial (along x axis) modulation of electric field (Fig. 4 (c)). In this case spatially modulated electric field takes the form:

$$E = E_{DC} + A\sin(qx). \tag{8}$$

It consists of constant part E_{DC} and of modulated part with amplitude $A \approx E_{DC}/10$. Simulations were performed for fixed values of temperature ($\xi/(k_BT) = 25$) and for strong ($\alpha/(k_BT) = 50$), middle ($\alpha/(k_BT) = 30$) and weak ($\alpha/(k_BT) = 10$) homogeneous anchoring energy. We have shown that in the cases of strong and medium anchoring regions of metamaterial and normal material are observed. When the anchoring is weak then real part of effective refractive index is negative almost in a whole sample.

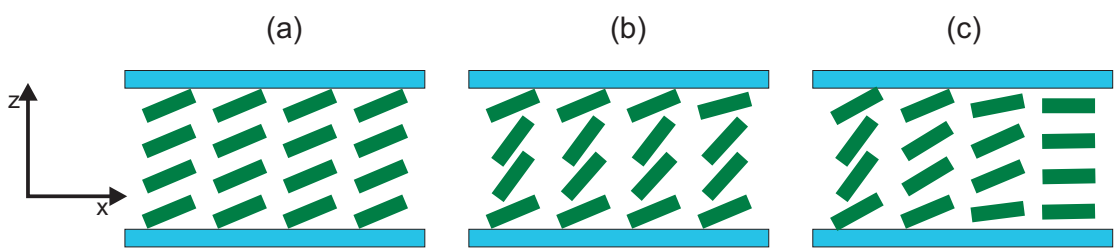

Figure 4. Schematicaly presented order in NLC cell: for global order(a), inhomogeneous order with constant electric field and anchoring (b) and inhomogeneous order with spatial modulation of electric field or anchoring forces.

The averaged index $\langle n'_{eff}\rangle(x)$, which is related to the phase shift of light transmitted through cell of thickness L along z-direction, was calculated according to formula:

$$\langle n'_{eff}\rangle(x) = \frac{1}{L}\int_0^L n'_{eff}(x,z)dz. \tag{9}$$

The amplitude of averaged index is related to diffraction efficiency η which is proportional to index amplitude squared:

$$\eta \propto (\Delta\langle n'_{eff}\rangle)^2. \tag{10}$$

We have shown[4] that the index amplitude modulation for system with spatial modulation of electric field is a few times larger than in a conventional LC system.

5. SPATIALLY MODULATED ANCHORING STRENGTH AND HOMOGENEOUS ELECTRIC FIELD

Inhomogeneity of anchoring can be as important as inhomogeneity of external electric field. Below, we investigate the simplest case when inhomogeneity is described by step function. Regions of surface with strong anchoring ($\alpha = 50$) are placed close to regions without anchoring ($\alpha = 0$) as shown schematically in Fig. 5. Anchoring

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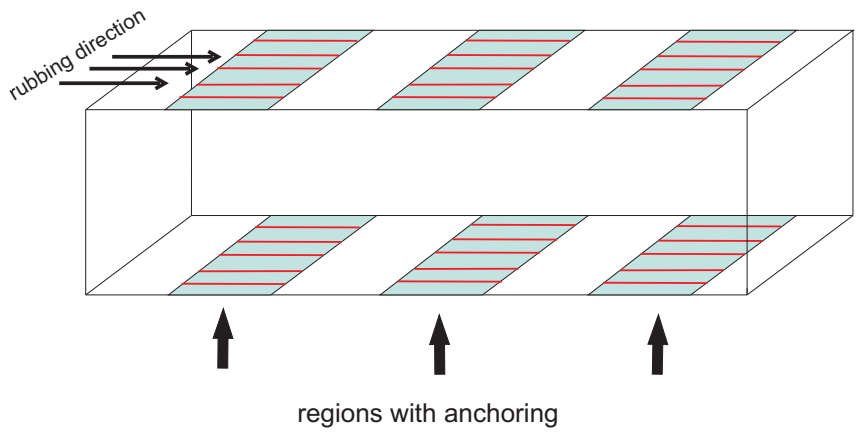

Figure 5. LC cell with modulation of anchoring.

energy reads:

$$\alpha(x) = \begin{cases} 50 & x \leq 50 \\ 0 & x > 50 \end{cases} .$$

Profile of anchoring energy $\alpha(x)$ for one period of modulation is presented in Fig. 6. In this section homogeneous electric field ($A = 0$) and constant temperature ($\xi/(k_B T) = 25$) are used.

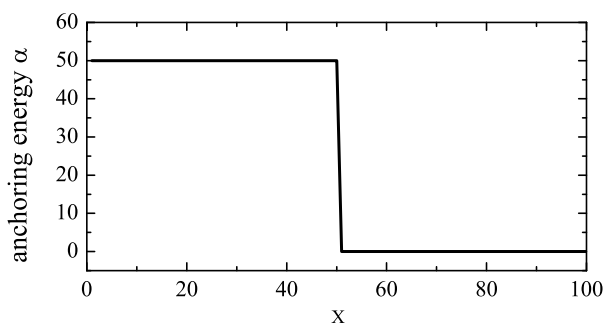

Figure 6. Profile of anchoring energy along x axis for one modulation period.

In Fig. 7 2D map of local refractive index for $\lambda = 2900$ nm and electric field $E = 0.75$ is presented. The case of $\lambda = 2820$ nm was briefly discussed in Ref.[13] Those preliminary results of simulations show that large difference between refractive index in neighboring places can be easily created in such systems. There are no horizontal (along x axis) layered structures which are typical of systems with modulated field. Presented preliminary results for $\lambda = 2900$ nm indicate that the index amplitude modulation for system with spatial modulation of anchoring forces is larger than in a system with modulation of electric field.

6. TEMPERATURE DEPENDENCE OF LOCAL DISTRIBUTION OF REFRACTIVE INDEX

The degree of order is characterized by order parameter, S, which takes values between 0 for a completely disordered phase and 1 for a completely ordered phase. Order parameter can be calculated from the second rank tensor:[2]

$$Q_{\alpha\beta} = \frac{1}{N}\sum_{i=1}^{N}(\frac{3}{2}\widehat{n}_{i,\alpha}\widehat{n}_{i,\beta} - \frac{1}{2}\delta_{\alpha\beta}),\ \alpha,\beta = x, y, z, \tag{11}$$

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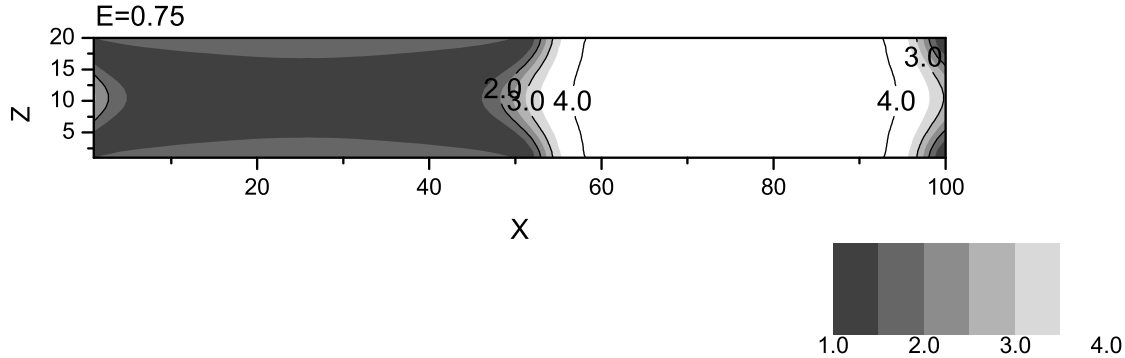

Figure 7. 2D maps of distribution of $n'_{eff}(x, y, z)$ averaged over y axis for $\lambda = 2900$ nm, $E = 0.75$.

where $\widehat{n}$ describes the orientation of the long axis of the LC molecule, N denotes number of molecules, and $\delta_{\alpha\beta}$ – the Kronecker delta function. Diagonalization of $Q_{\alpha\beta}$ gives three eigenvalues λ_+, λ_0 and λ_- ($\lambda_+ > \lambda_0 \geq \lambda_-$) which sum up to zero. For an uniaxial system the eigenvalues satisfy the relation:

$$\lambda_- = \lambda_0 = -\frac{1}{2}\lambda_+. \tag{12}$$

In this paper the order parameter $S = \lambda_+$ is used.

Monte Carlo simulations were done for reduced value of critical temperature $k_BT/\xi = 1.15$. The presence of field and anchoring may have a strong impact on the character of the transition - this topic will be studied in subsequent papers. In Fig. 8 2D maps of local refractive index for low temperature ($k_BT/\xi = 0.4$, $S \approx 0.9$) and high temperature ($k_BT/\xi = 1.12$, $S \approx 0.4$), which is close to the critical temperature of pure NLC system, are shown. We have used the wavelength $\lambda = 2820$ nm and $E_{DC} = 1.2$. In the middle of the sample for low temperature the refractive index is negative and lower then -0.5. For higher temperature this value increases. Thus, the temperature can be an additional tuning parameter for NDLC system.

7. CONCLUSIONS

We have presented a quantitative analysis of the effect of an applied spatially modulated field and spatially modulated anchoring energy on a metamaterial comprising nano-spheres dispersed in an aligned nematic liquid crystal, using recent and new Monte Carlo simulations. The first results indicate that amplitude of refractive index can be much larger than that for a conventional LC system. Additionally, we have shown that the metamaterial properties of NDLC are temperature dependent.

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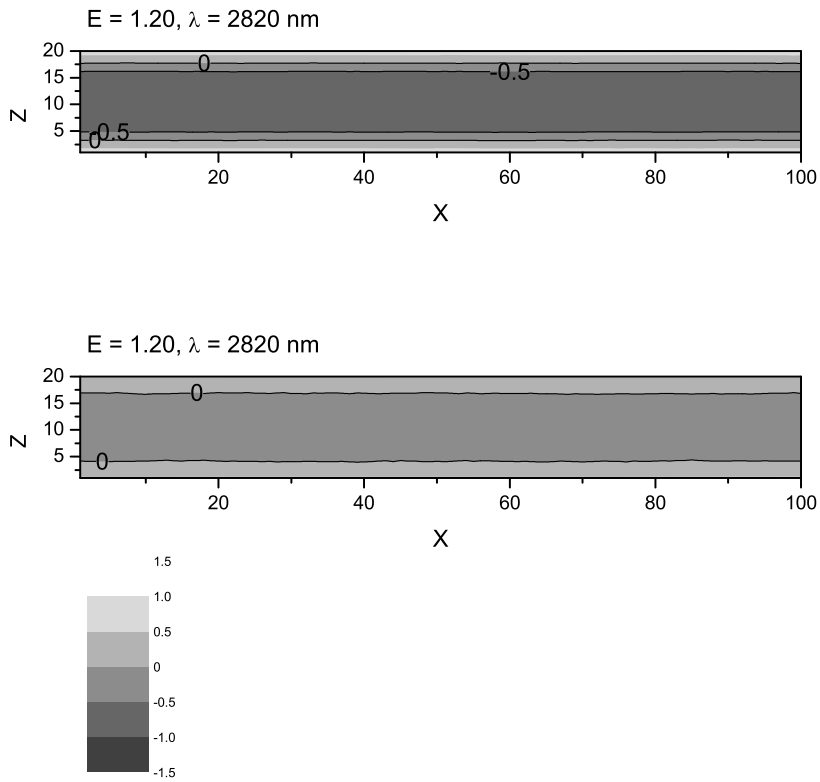

Figure 8. 2D maps of local refractive index for low ($k_BT/\xi = 0.4$, top) and high ($k_BT/\xi = 1.12$, bottom) temperature.

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